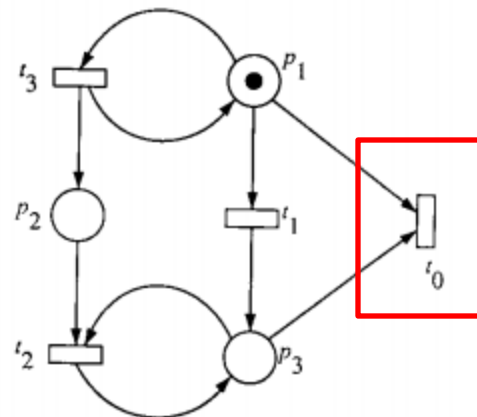
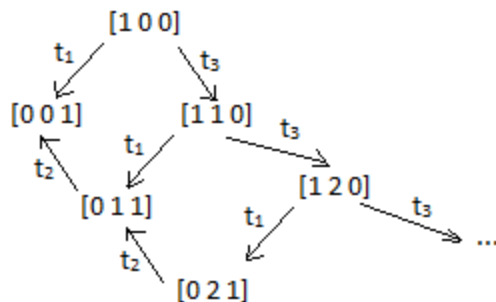


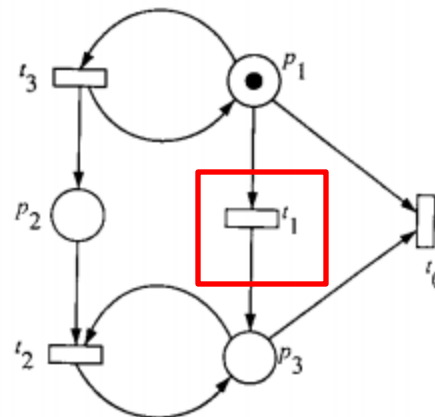
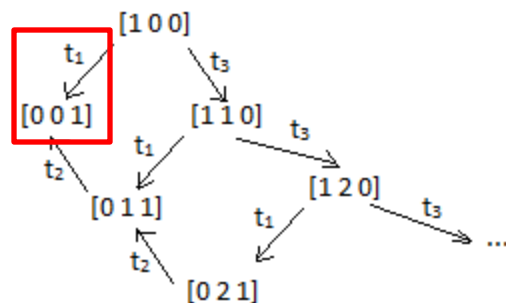
Petri Net Liveness

- **L0-Live (dead):** a particular transition can never fire
 - In the figure below, t_0 can never fire (see reachability graph)



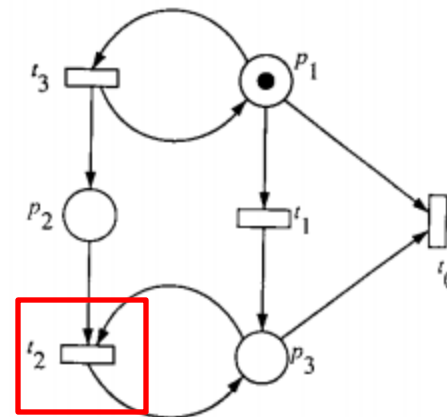
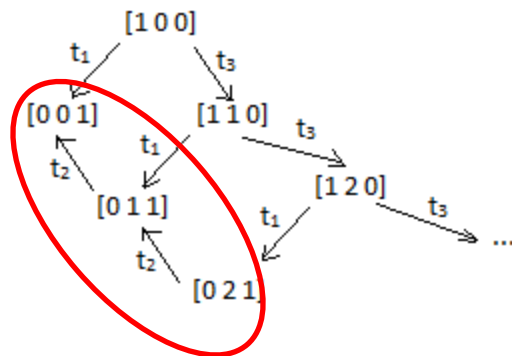
Petri Net Liveness

- **L1-Live:** a particular transition can fire at least once for any firing sequence
 - In the figure below, t_1 can only fire once, for any firing sequence (see reachability graph)



Petri Net Liveness

- **L2-Live:** a particular transition can fire k times for a particular firing sequence, for any k .
 - In the figure below, t_2 can only fire once, twice, thrice, etc, for different firing sequences



Petri Net Liveness

- **L3-Live:** a particular transition can fire infinitely in a particular firing sequence.
 - In the figure below, t_3 can fire infinitely for the firing sequence $t_3, t_3, t_3, t_3, \dots$
 - Note that the number of times t_1 and t_2 fire is finite for any firing sequence.

